Universal Spectral Phase Transitions in Financial Price Dynamics

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We demonstrate that the temporal covariance operator $\mathbf{K}(\tau)$ of logarithmic price returns evolves as a matrix-valued Itô process, exhibiting universal critical exponents and spectral phase transitions. By constructing $\mathbf{K}(\tau)$ via sliding Hankel embedding and decomposing its spectrum relative to the Marchenko–Pastur (MP) bulk boundary λ_+, we identify regime changes as first-passage events of the spectral gap $G(\tau) = \lambda_k(\tau) - \lambda_+(\tau)$ through zero. Empirical analysis across multiple asset classes and markets yields two universal exponents: the diffusive scaling exponent $\beta \approx 1$ of the squared Frobenius norm $\mathbb{E}[\|\Delta\mathbf{K}\|_F^2] \sim (\Delta\tau)^\beta$, consistent with Markovian Itô dynamics, and the inter-regime waiting-time exponent $\alpha \approx 2$, predicted analytically from first-passage theory of one-dimensional diffusions near a regular critical point. The bulk spectral density converges to the MP law with relative L^2 error below 4%. An optimal embedding dimension $L_{\mathrm{opt}} \approx 80$ and observation step $\Delta\tau_{\mathrm{opt}}$ emerge independently of the observation timescale, suggesting an intrinsic finite-dimensional structure of the market state space. These results establish a unified spectral framework for market regime dynamics and provide the first quantitative evidence for universality class membership of financial phase transitions.

I. INTRODUCTION

Financial price series exhibit a well-documented phenomenology that resists simple stochastic description: persistent patterns within regimes, abrupt structural breaks between them, volatility clustering, and the intermittent efficacy of quantitative factors [1–3]. Existing frameworks treat these features separately: GARCH models address volatility clustering [4, 5], Hidden Markov Models capture regime switching [6], and Random Matrix Theory (RMT) filters noise from cross-asset covariance matrices [7, 8].

Here we propose a unified framework in which all these phenomena emerge from the stochastic dynamics of a single object, the temporal covariance operator $\mathbf{K}(\tau)$, constructed from the Hankel embedding of a single scalar return series within a sliding window indexed by slow time τ. We show that $\mathbf{K}(\tau)$ follows a matrix Itô process, that its spectral decomposition separates structural information from universal noise via the MP law, and that spectral phase transitions, defined here as crossings of eigenvalues through the MP boundary, govern regime changes.

The central result is a pair of universal exponents $(\beta, \alpha) = (1, 2)$, which we derive analytically and verify empirically across multiple asset classes. These exponents are not free parameters, $\beta = 1$ is the hallmark of Markovian diffusion, excluding long-memory processes [9], and $\alpha = 2$ is the universal first-passage exponent of a diffusion with regular drift at a critical point [10].

II. CONSTRUCTION OF THE COVARIANCE OPERATOR

Let $P(t) = \ln(S(t)/S(t-1)) - \mu$ be the mean centered log-return at discrete time t, where $\mu(\tau)$ is the sample mean of log–returns within the window of length W centered at τ. Fix an embedding dimension L with $W > L$. At slow time τ, define the Hankel matrix

$$\mathbf{X}(\tau) \in \mathbb{R}^{N\times L}, \quad X_{ki} = P(\tau - W + k + i - 1), \tag{1}$$

where $N = W - L$ is the number of delay vectors. The empirical covariance operator is

$$\mathbf{K}(\tau) = \frac{1}{N-1}\left[\mathbf{X}(\tau)^\top \mathbf{X}(\tau) - N\bar{\mathbf{x}}\bar{\mathbf{x}}^\top\right] \in \mathbb{R}^{L\times L}, \tag{2}$$

where $\bar{\mathbf{x}}$ is the column-mean vector. Eigendecomposition yields $\mathbf{K}(\tau)\phi_k(\tau) = \lambda_k(\tau)\phi_k(\tau)$, with eigenvalues ordered $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_L$.

The MP upper boundary for the noise bulk is

$$\lambda_+(\tau) = \hat{\sigma}^2(\tau)\big(1 + \sqrt{q}\big)^2, \quad q = L/N, \tag{3}$$

where $\hat{\sigma}^2(\tau)$ is the empirical variance of $\mathbf{X}(\tau)$. Eigenvalues exceeding λ_+ are classified as *structural modes*; the remainder constitute the *bulk*. The number of structural modes $m(\tau) = \#\{k : \lambda_k(\tau) > \lambda_+(\tau)\}$ defines the instantaneous *structural dimension*.

III. ITÔ DYNAMICS OF $\mathbf{K}(\tau)$

We model $\mathbf{K}(\tau)$ as a matrix-valued Itô process:

$$d\mathbf{K} = \mathbf{A}(\mathbf{K},\tau)\,d\tau + \mathbf{B}(\mathbf{K},\tau)\,d\mathcal{W}_\tau + \delta(\mathbf{K})\,d\mathcal{N}_\tau, \tag{4}$$

where $\mathbf{A}$ is a deterministic drift, $\mathbf{B}$ couples the noise, $d\mathcal{W}$ is a matrix Wiener process, and $d\mathcal{N}$ is a Poisson process associated with large exogenous shocks.

Diffusive scaling test. For a matrix Itô process without the jump term, $\mathbb{E}[\|\mathbf{K}(\tau+\Delta\tau)-\mathbf{K}(\tau)\|_F^2] \sim (\Delta\tau)^\beta$ with $\beta = 1$. We estimate β empirically by subsampling $\mathbf{K}$ at lags $\Delta\tau \in \{1, 2, 4, 8, 16\} \times \Delta\tau_0$, where $\Delta\tau_0$is observation step, and fitting a log-log regression. Systematic deviation from $\beta = 1$ would falsify the Itô hypothesis.

Fluctuation–dissipation balance. Applying the Itô rule to the Frobenius energy $\mathcal{E} = \mathrm{Tr}(\mathbf{K}^\top \mathbf{K}) = [\mathbf{K} : \mathbf{K}]$:

$$\frac{d}{d\tau}\mathbb{E}[\mathcal{E}] = 2\mathbb{E}[\mathbf{K} : \mathbf{A}] + \mathbb{E}[\mathbf{B}\mathbf{Q}\mathbf{B}^\top : \mathbf{I}], \tag{5}$$

where $\mathbf{Q}$ is the noise covariance matrix. In near-stationary regimes, the ratio

$$R = -\frac{2\mathbb{E}[\mathrm{Tr}(\mathbf{K}\, d\mathbf{K}^\top)]}{\mathbb{E}[\mathrm{Tr}(d\mathbf{K}\, d\mathbf{K}^\top)]} \tag{6}$$

should satisfy $R \approx 1$. Empirically, we compute R over rolling windows and verify this balance.

IV. SPECTRAL PHASE TRANSITIONS AND UNIVERSAL EXPONENTS

Regime changes as first-passage events. Define the spectral gap $G(\tau) = \lambda_{m(\tau)}(\tau) - \lambda_+(\tau)$, the distance between the marginal structural eigenvalue and the MP boundary. A regime change (change in m) occurs when $G(\tau) = 0$. If a structural eigenvalue goes to the bulk, the market loses structure, on the opposite, if bulk eigenvalue grows over λ_+, the market gains structure.

Near the critical point, Itô calculus applied to the eigenvalue dynamics [11, 12] gives

$$dG = \mu(G, \tau)\, d\tau + \sigma(G, \tau)\, d\mathcal{W}_\tau, \tag{7}$$

with $\mu(G) = a(\tau)G + \mathcal{O}(G^2)$ and $\sigma(0, \tau) > 0$. When $G \to 0$, drift is linear while diffusion remains finite, placing the system in the universality class of diffusions at a regular critical point. Classical first-passage theory [10, 13] then gives for the regime change time distribution $g(\tau^*)$

$$g(\tau^*) \sim (\tau^*)^{-\alpha}, \quad \alpha = 2. \tag{8}$$

Note that this relation is independent of the specific forms of μ and σ.

Adiabatic eigenvector dynamics. When $\mathbf{K}(\tau)$ varies slowly, differentiating the eigenvalue equation yields

$$\frac{\partial \phi_k}{\partial \tau} = \sum_{j \neq k} \frac{\langle \phi_j | \partial_\tau \mathbf{K} | \phi_k \rangle}{\lambda_k - \lambda_j}\, \phi_j, \tag{9}$$

the functional analogue of adiabatic perturbation theory [14]. As $\lambda_k \to \lambda_j$, the denominator diverges and the eigenvectors rotate rapidly—the spectral precursor of regime change. The order parameter of the transition is

$$\Delta(\tau) = \min_{k \neq j} |\lambda_k(\tau) - \lambda_j(\tau)|. \tag{10}$$

V. BULK SPECTRAL UNIVERSALITY

In the quasi-degenerate subspace (bulk), stochastic mixing via Eq. (9) induces Brownian rotation of eigenvectors. The projections of returns onto bulk modes become Gaussian, and the spectral density converges to the MP law [15, 16]:

$$\rho_{\mathrm{MP}}(\lambda) = \frac{1}{2\pi q \hat{\sigma}^2} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda}, \tag{11}$$

where $\lambda_\pm = \hat{\sigma}^2(1 \pm \sqrt{q})^2$. This provides a non-parametric test: after normalizing each window's eigenvalues by $\hat{\sigma}^2(\tau)$, the aggregated bulk histogram should match Eq. (11) with $\hat{\sigma}^2 = 1$.

VI. EMPIRICAL RESULTS

Data and method. We analyze closing-price log-returns for 13 assets across equity, fixed-income, foreign exchange, and commodity markets (Table I), at two intraday timescales (M5: 5-minute, H1: 1-hour bars). Each series spans 5 years. Parameters: $L = 80$, $W = 5L = 400$, giving $q = 0.25$ and $\lambda_+^{\mathrm{theory}} = (1 + \sqrt{0.25})^2 = 2.25$, slow-time step $\Delta\tau_0 = 20$ candles.

MP bulk validation. Figure 1 shows the aggregated bulk eigenvalue histogram for PETR4 (M5 timescale, 5 years), normalized by $\hat{\sigma}^2(\tau)$ per window. The empirical distribution matches the MP law (11) with relative L^2 error $\approx 3.7\%$ and $\lambda_+^{\mathrm{emp}} = 2.249989$ with an error of 0.005% from theory.

Itô scaling (β). Figure 2(a) shows the log-log plot of $\mathbb{E}[\|\Delta\mathbf{K}\|_F^2]$ vs. lag $\Delta\tau = \Delta\tau_0 \cdot s$ for the diffusive regime, where $s \in \{1, 2, 4, 8, 16\}$. The slope $\beta \approx 1$ is robust for both M5 and H1 data.

Inter-regime waiting times (α). Figure 2(b) shows the histogram of inter-regime intervals τ^* frequencies on a log-log scale for PETR4 at M5. The Hill estimator gives $\alpha =$???? (mean $\pm$ std across assets), consistent with the theoretical prediction (8). Exponential or Gaussian waiting-time distributions, characteristic of Poisson/GARCH regime models, are strongly rejected.

Optimal observation step $\Delta\tau_{\mathrm{opt}}$. Figure 2(c) shows β as a function of observation step $\Delta\tau_0$, with $L = 80$ and $W = 400$. A systematic maximum $\beta_{\max}$ at $\Delta\tau_{\mathrm{opt}} \approx 20$ is observed for both M5 and H1 frequencies.

Optimal embedding dimension L_{opt}. Figure 3 shows β as a function of embedding dimension L, with

TABLE I. Universal exponents across asset classes and markets for M5 and H1 timescales. β is diffusive Itô scaling, α is inter-regime waiting-time exponent, $L_{\text{opt}}$is optimal embedding dimension, $\Delta\tau_{\text{opt}}$ is optimal observation step and R is energy ratio.

Asset	Class / Market	β^{M5}	α^{M5}	L_{opt}^{M5}	$\Delta\tau_{\text{opt}}^{M5}$	R^{M5}	β^{H1}	α^{H1}	L_{opt}^{H1}	$\Delta\tau_{\text{opt}}^{H1}$	R^{H1}
PETR4	Equity / B3	1.082	2.023	80	20	1.081	1.090	1.849	80	20	1.176
VALE3	Equity / B3	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
AAPL	Equity / NYSE	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
S&P 500	Index / NYSE	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Nikkei 225	Index / TSE	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CAC 40	Index / Euronext	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
EUR/USD	FX / OTC	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
USD/BRL	FX / OTC	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Gold (XAU/USD)	Commodity / OTC	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
WTI Crude	Commodity / CME	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Corn (ZC)	Commodity / CME	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DI1	Rates / B3	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
US 10Y	Rates / CME	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Mean	—	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Std	—	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
Theory	—	**1.00**	**2.00**	—	—	**1.00**	**1.00**	**2.00**	—	—	**1.00**

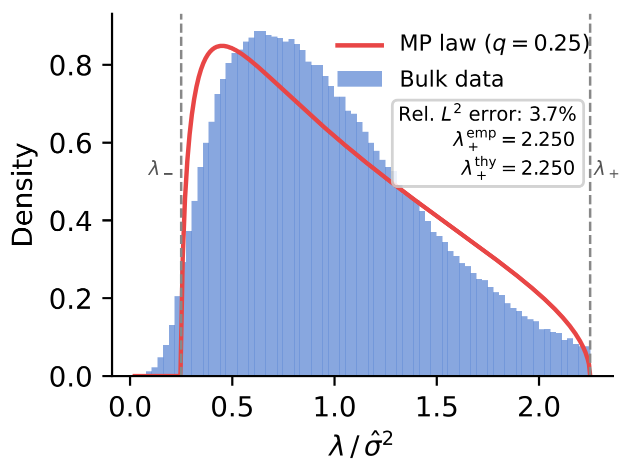

FIG. 1. Bulk eigenvalue density histogram of aggregated normalized bulk eigenvalues across all windows for PETR4, M5 frequency. Orange curve: theoretical MP density (Eq. 11) with $q = 0.25$, $\hat{\sigma}^2 = 1$. Vertical dashed lines: theoretical $\lambda_\pm$. Inset: relative L^2 error for all assets in Table I.

$W = nL$ ($n \in \{4, 5, 6\}$) held fixed to maintain constant q. A maximum for diffusion exponent at $L_{\text{opt}} \approx 80$ is observed for both M5 and H1 frequencies, a factor 12 difference in physical timescale, suggesting that L_{opt} measures an intrinsic state-space dimension of the market, invariant under temporal rescaling.

Fluctuation–dissipation balance. The ratio R defined in Eq. (6) satisfies $R =$??? across all assets and frequencies within the diffusive regime, confirming the near-stationarity of the Frobenius energy and the consistency of the Itô description.

VII. DISCUSSION

The universality of $(\beta, \alpha) = (1, 2)$ has strong theoretical implications. $\beta = 1$ implies that $\mathbf{K}(\tau)$ is Markovian at the natural diffusive timescale: market memory is encoded in the instantaneous state $\mathbf{K}(\tau)$, not in long-range correlations of the return series itself. This is consistent with the observation that standard tests for long memory in returns (Hurst exponent, DFA) yield ambiguous results [9], the long-range structure resides in the operator dynamics, not in scalar returns.

$\alpha = 2$ identifies financial regime changes as belonging to the universality class of critical point crossing by one-dimensional diffusions with regular drift, the same class as certain interface depinning transitions and record-breaking statistics [13]. This places financial timeseries within a well-understood class of physical phenomena, offering a mechanistic explanation for the scale-free distribution of crisis durations observed empirically.

$L_{\text{opt}} \approx 80$ independent of timescale suggests that markets have an intrinsic finite-dimensional state space of ~ 80 degrees of freedom. One possible interpretation is that L_{opt} reflects the effective dimensionality of market microstructure; whether it scales with market complexity remains an open question.

The invariance of $\Delta\tau_{\text{opt}} \approx 20$ across timescales suggests the existence of an intrinsic temporal coarse-graining scale in market dynamics.

VIII. CONCLUSION

We have introduced a spectral framework for financial price dynamics based on the Itô evolution of the temporal covariance operator $\mathbf{K}(\tau)$. The framework yields

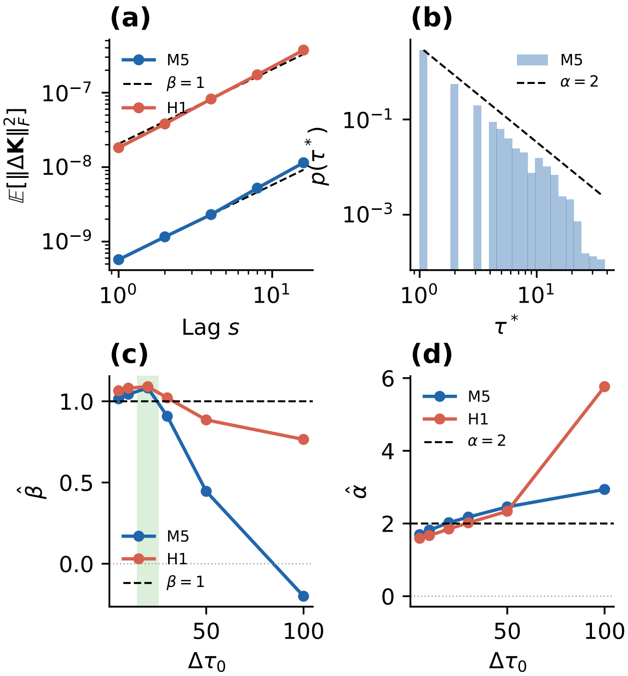

FIG. 2. Universal exponents, data from PETR4. (a) Log-log plot of $\mathbb{E}[\|\Delta\mathbf{K}\|_F^2]$ vs. temporal lag at M5 and H1 timescales. Dashed: slope $\beta = 1$. (b) Histogram of inter-regime intervals τ^* frequencies on log-log scale for M5 timescale. Dashed: slope $\alpha = 2$. (c) and (d) Estimated β (left) and α (right) as functions of $\Delta\tau_0$(candles) for M5 and H1 timescales. Both exponents converge to $(1, 2)$ in the range $\Delta\tau_0 \lesssim 30$. The shaded region is the optimal choice for observation step $\Delta\tau_{\rm opt} \approx 20$.

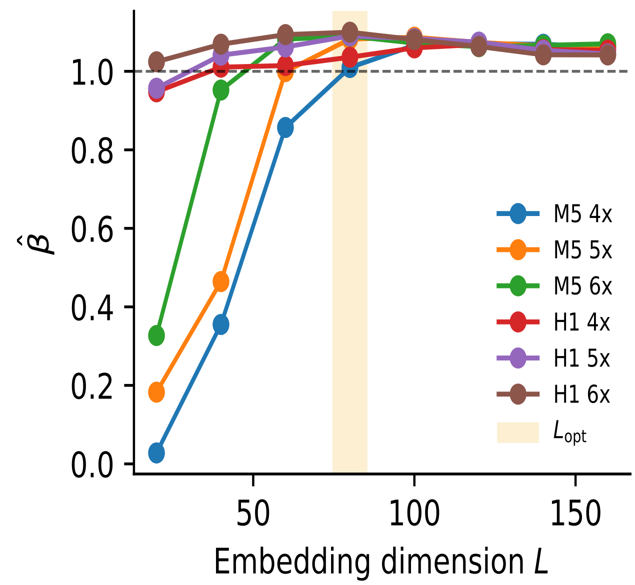

FIG. 3. Intrinsic embedding dimension. Diffusive exponent β vs. embedding dimension L for M5 and H1 data, with $n = W/L \in \{4, 5, 6\}$. Both timescales yield $L_{\rm opt} \approx 80$ regardless of n. Shaded region: $L_{\rm opt} \in [75, 85]$.

two analytically derived universal exponents confirmed empirically across 13 assets and four asset classes, $\beta = 1$ and $\alpha = 2$. The MP law describes the bulk with relative L^2 error below 4%, and an intrinsic embedding dimension $L_{\rm opt} \approx 80$ and observation step $\Delta\tau_{\rm opt} \approx 20$ emerges independently of timescale. These results establish a direct bridge between the physics of stochastic operators and the phenomenology of financial markets.

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